24/2/26

Recitation: 5

An email service provider wants to classify emails as Spam / Not Spam using a linear SVM classifier.

Spam = +1 , Not Spam = -1

Suspicious words	links	length	Attachment	Class
6	3	100	1	+1
1	0	350	0	-1
4	2	150	1	+1
0	0	420	0	-1
5	2	120	1	+1

$$f(x) = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4$$

$$y_i (x_1 w_1 + x_2 w_2 + w_3 x_3 + w_4 x_4) \geq 1$$

$$y_i (x_i \cdot w_i + b) \geq 1$$

$$\text{margin} = \frac{2}{||w||}$$

$$||w|| = \sqrt{w_1^2 + w_2^2 + w_3^2 + w_4^2}$$

if $y_i (w \cdot x + b) \geq 1$ Perfect classification

if < 1 - Classification failed

update weights

if Perfect

$$w_i^{New} = w_i^{old} - \eta w_i^{old}$$

if Wrong

$$w_i^{New} = (1-\eta) w + \eta \sum y_i x_i$$

$$f(x) = 0 \Rightarrow w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$$

	SW	L	A
Spam	(4, 5, 6)	(2, 3)	(1)
Non Spam	(0, 1)	(0)	(0)

$w_2, w_3 = 0$

$x_1 = 2, \quad w = 1$

$w_0 + w_1 x_1 = 0$

$w_0 + 2 = 0$

$w_0 = (-2)$

Check $\quad y_i(x_i - 2)$

$= 1(6-2) = 4 > 1$

$= -1(4-2) = 1 \geq 1$

$= 1(4-2) = 2 \geq 1$

$= 1(4-2) = 2 \geq 1$

$-1(0-2) = 2 \geq 1$, $1(5-2) = 3 \geq 1$

$\therefore$ All classification are perfect.

Loss function:

$$L = \frac{1}{2}\|w\|^2 - \sum_{i=1}^{N} C_i\left[y_i(w x_i + b) - 1\right]$$

$$= \frac{1}{2}(w_1^2 + w_2^2 + w_3^2) - \sum_{i=1}^{N} C_i \text{ Error}_i$$

~~To~~ PF > $y_i(\Sigma w_i x_i) \geq 1$

$C \sim 0$, $J = \frac{1}{2}w^2$

$$\frac{dJ}{dw} = w$$

$$w_1^{New} = w_1^{old} - \eta w_1^{old}$$

$$= 1 - 0.1(1) = 0.9$$

$$m = \frac{2}{\|w\|} = \frac{2}{\sqrt{(0.9)^2 + 0^2 + 0^2}} = 2.222$$

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~~If N.P.F~~

~~$y_i(w_1x_1 + w_2x_2) < 1$ take~~

$$\frac{2}{}$$

$$w_{new} = w - \eta \frac{dL}{dw}$$

$$w^{new} = w - \eta \frac{dL}{dw} = \frac{2}{0.9} = 2.222$$

$$w - \eta \left[w - \sum c_i y_i x_i \right)$$

$$(1-n)\, w + \eta \sum_i c_i\, y_i x_i \Big)$$

$$(=1=) \; (y_1x_1 + y_2x_2 + y_3x_3 + y_4x_4 + y_5x_5)$$

$$= (1)(6) + (-1)(1) + (1)4 - 1(0) + 1(5)$$

$$b_{new} = b^{old}_{SVM} - \eta \frac{dL}{db}$$

$$= b_0^{1d} - \eta \left[\sum c_i y_i \right]$$

$$w_{i\,new} = -b$$

$$w_2 = -3$$

$$w_3 = -1$$

$$b = 1$$

$y_i f(x) = +5 \geq$ Perfect classification

but also update the weights.

$w = w - \eta \cdot (w - yx)$

$w_1^{New} = w_1^{old} - \eta \frac{dL}{dw}$

If right Prediction $\underset{\text{Lagrangian}}{L} = \frac{1}{2} \|w\|^2$

$w_1^{old} - \eta w_1^{old}$

$w_1^{new} = -6 - 1(-6) = 0$

$w_2^{new} = -3 - 1(-3) = 0$

$w_3^{New} = 0$ $\qquad b = b + \eta y$

$L = \frac{1}{2} \|w\|^2$

$\text{margin} = \frac{2}{\|w\|} = \frac{2}{\sqrt{w_1^2 + w_2^2 + w_3^2}}$

$b^{New} = b^{old} - \eta \frac{dL}{dw}$

$b^{New} = b^{old} - 1$